Regulatory Reform and the Composition of FDA Drug Approvals

Evidence on Controlled Substances, Regulatory Incentives, and Pharmaceutical Market Composition

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*

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The Prescription Drug User Fee Act (PDUFA) of 1992 authorized the FDA to collect industry user fees in exchange for performance targets on review times. While a large literature documents that PDUFA substantially accelerated drug approval, no study has asked whether regulatory acceleration also changed the *composition* of approved drugs—specifically, whether the share of approvals classified as controlled substances under the Controlled Substances Act changed as a result. Using the Drugs@FDA database linked to DEA scheduling records (1939–2025, $N = 25{,}908$ original approvals), I estimate interrupted time series and difference-in-differences specifications around both PDUFA (1992) and the Generic Drug User Fee Amendments (GDUFA, 2012). The overall controlled-substance share of FDA approvals fell after PDUFA. Focusing on new drug applications (NDAs), where the policy incentive is most direct, a Poisson rate model yields an incidence-rate ratio of 1.46 ($p = 0.06$) that is not robust to group-specific trend controls. A late-2000s spike in Schedule II NDA approvals (OxyContin reformulation, Exalgo, Embeda, and related products) accounts for the apparent post-PDUFA rate increase. For GDUFA, the ANDA controlled-substance share fell significantly post-reform ($\hat{\beta} = -0.40$, IRR $= 0.67$, $p < 0.001$). The evidence does not support the hypothesis that user-fee-funded regulatory acceleration systematically increased the controlled-substance share of FDA approvals.

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1. Introduction

Drug overdose deaths reached 107,000 in the United States in 2021, the highest annual total on record, and prescription controlled substances have played a prominent role throughout the overdose epidemic (Mark and Huber 2024). Understanding how regulatory institutions shape the legal supply of controlled substances is therefore a question with first-order policy consequences.

The FDA is the primary gatekeeper of the U.S. pharmaceutical market. Its decisions determine which drugs may be sold, how quickly they reach patients, and—through the review process—what evidence of safety and efficacy is required before approval (Carpenter 2010). A large literature examines the speed-safety tradeoff at the FDA: faster review times accelerate patient access to beneficial new treatments, but deadline pressure may also reduce the scrutiny applied to drugs with later safety problems (Carpenter, Zucker, and Avorn 2008; Philipson et al. 2008). The Prescription Drug User Fee Act (PDUFA) of 1992 stands at the center of this literature: by authorizing the FDA to collect manufacturer fees in exchange for binding review-time targets, PDUFA substantially cut median approval times and changed the incentive environment facing both the agency and the firms it regulates (Ernst R. Berndt et al. 2005; M. Olson 2009).

A parallel literature studies the economics of controlled substances. Pharmaceutical opioids, stimulants, benzodiazepines, and other DEA-scheduled drugs are clinically important but carry non-trivial abuse potential. Supply-side interventions in these markets can have unintended consequences: Alpert, Powell, and Pacula (2018) show that abuse-deterrent OxyContin reformulation reduced prescription misuse but shifted users toward heroin, illustrating how the composition of the legal drug supply feeds upstream into illicit-market behavior.

What no study has examined is whether PDUFA—or its generic-drug analogue, the Generic Drug User Fee Amendments (GDUFA) of 2012—changed the composition of FDA approvals in ways that matter for controlled-substance supply. If review-time incentives disproportionately benefit high-margin drug classes, and if controlled substances systematically generate higher margins through patent-protected branded formulations, then regulatory acceleration may have altered the mix of approved drugs even without any deliberate policy intent.

This paper provides the first systematic test of that hypothesis. Using the complete Drugs@FDA database (1939–2025, $N = 25{,}908$ original approvals) linked to DEA scheduling records, I estimate interrupted time series and difference-in-differences specifications around both PDUFA (1992) and GDUFA (2012). The central finding is a null result with an important qualification. The aggregate controlled-substance share of FDA approvals fell after PDUFA, from 9.6% pre-1992 to 7.3% post-1992. A Poisson rate model restricted to new drug applications—where the PDUFA incentive is most direct—yields an incidence-rate ratio of 1.46 that is marginally significant ($p = 0.06$) under year fixed effects but collapses to 1.19 ($p = 0.62$) once group-specific time trends are added. A drug-level investigation of the 2009–2013 CS-NDA spike that drives the apparent effect reveals it to be an industry opioid product cycle (OxyContin reformulation, Exalgo, Embeda, Suboxone, and related extended-release and abuse-deterrent formulations), not a PDUFA composition effect. For GDUFA, the ANDA controlled-substance share declined significantly in the post-performance-goal era (IRR = 0.67, $p < 0.001$), which is inconsistent with the regulatory acceleration hypothesis for generic drugs.

The broader implication is that the speed-safety tradeoff literature should be extended to

downstream market-composition questions with care: the controlled-substance composition of FDA approvals is not stable over time, but the variation appears to reflect industry product cycles rather than regulatory incentives.

Roadmap. Section 2 reviews the institutional background and related literature. Section 3 describes the data and variable construction. Section 4 presents the empirical strategy. Section 5 reports the results. Section 6 interprets the findings and discusses limitations. Section 7 concludes.

2. Background and Related Literature

2.1. Institutional Background

PDUFA (1992). The Prescription Drug User Fee Act, enacted in October 1992, authorized the FDA to collect fees from pharmaceutical manufacturers to supplement congressional appropriations for drug review (Rep. Dingell 1992). In exchange, the agency agreed to binding performance goals specifying median review times for new drug applications. The original PDUFA I (1993–1997) committed the FDA to reviewing standard applications within 12 months and priority applications within 6 months of receipt, with completion targets phased in over five years. Subsequent reauthorizations—PDUFA II through VII—have progressively tightened review-time targets, expanded the user-fee program to biologics, and added post-market safety obligations. The fee structure is negotiated between the pharmaceutical industry and the FDA before each five-year reauthorization cycle, a feature that has raised concerns about regulatory capture (Darrow, Avorn, and Kesselheim 2017).

PDUFA applies to applications for new molecular entities and new formulations submitted under the NDA pathway. It does not directly govern the Abbreviated New Drug Application (ANDA) pathway for generic drugs, which was created by the Hatch–Waxman Act of 1984 (Rep. Waxman 1984). Before 2012, ANDA reviews were funded entirely by congressional appropriations, which contributed to a backlog of thousands of pending generic applications.

GDUFA (2012). The Generic Drug User Fee Amendments, signed in July 2012, extended the user-fee model to generic drug reviews under the ANDA pathway (Research Wed, 05/08/2024 - 11:53). GDUFA I (2013–2017) collected fees from generic manufacturers and established performance goals for ANDA review times. A distinguishing feature of the initial GDUFA implementation was a substantial transition period (2013–2014) during which the FDA worked to reduce a pre-existing backlog of approximately 2,500 unreviewed applications; the performance-goal era for new ANDA submissions effectively began in 2015. GDUFA II (2018–2022) and GDUFA III (2023–2027) have further refined the fee structure and review targets. Because 86.6% of controlled-substance original approvals in this dataset come through the ANDA pathway, GDUFA is arguably the more relevant policy for this paper's central question.

2.2. Theory of User-Fee Incentives

The standard economic argument for why PDUFA might alter drug composition runs through the profit motive. Under a user-fee regime with binding review-time targets, the FDA faces stronger incentives to process applications quickly. If faster review reduces the expected approval lag, it increases the net present value of approval for any drug. However, the value of accelerated approval should be highest for drugs with the largest expected revenue streams, because a shorter time-to-market compresses the revenue-loss period before patent expiration.

Controlled substances—particularly branded opioid analgesics—occupy a distinctive position in the pharmaceutical market. Many schedule II opioid formulations carry high list prices, face limited generic competition during patent protection, and generate substantial revenue for innovator firms (Powell, Pacula, and Taylor 2020). If the value of accelerated review is proportional to expected product revenue, then PDUFA may have increased the relative attractiveness of developing controlled-substance NDAs compared with other pharmaceutical product classes.

A competing hypothesis holds that the primary effect of PDUFA was to increase FDA *staffing and review resources* rather than to change the incentive environment for drug developers directly. Carpenter, Smith, et al. (2026) argue that the source of FDA funding—user fees versus congressional appropriations—matters less than the total level of review resources. Under this view, PDUFA improved review-time performance primarily by enabling the FDA to hire additional staff and retain expertise, and any composition effect would operate through the supply of review capacity rather than through demand-side incentives for manufacturers.

2.3. Related Empirical Literature

PDUFA, review speed, and approval. The most direct predecessor is Ernst R. Berndt et al. (2005), who use drug-level panel data to show that PDUFA substantially reduced NDA review times and increased approval rates, while finding little evidence of increased post-approval withdrawals. M. Olson (2009) examines launch decisions and finds that PDUFA increased the probability of U.S. initial launch conditional on development, consistent with the thesis that faster review improves the expected return to development. M. K. Olson (1997) documents substantial heterogeneity in pre-PDUFA review times across firm types, suggesting that any compositional effects may depend on which types of firms were most constrained by review delays before 1992. Dranove and Meltzer (1994) provide an early analysis of whether important drugs reach the market faster, finding limited evidence of differential queue position by therapeutic value.

Post-market safety. Carpenter, Zucker, and Avorn (2008) exploit the feature that drugs approved just before PDUFA deadlines received less review time than those approved just after, finding that proximity-to-deadline approvals are associated with higher rates of subsequent safety-related labeling changes. Darrow, Avorn, and Kesselheim (2020) provides a broad review of the literature and argues that expedited pathways have reduced clinical evidence requirements at the time of approval, raising concerns about downstream safety for certain product classes. Both

papers are informative about the speed-safety margin but do not address the market-composition question examined here.

Welfare analysis. Philipson et al. (2008) develop a formal welfare model of the FDA approval decision and use it to estimate that approval delays under the pre-PDUFA regime imposed large welfare costs. The framework emphasizes the consumer-surplus value of earlier access to beneficial drugs but does not model externalities associated with abuse potential or diversion. That omission is consequential for controlled substances, where the social cost of approval may include downstream misuse spillovers that are not internalized by approving firms or patients.

Generic drugs and GDUFA. Ernst R Berndt, Conti, and Murphy (2018) provide the first systematic economic analysis of GDUFA, documenting that the reform reduced ANDA review times and approval lags. They also analyze barriers to generic entry and argue that regulatory delays were a significant constraint on competition in the pre-GDUFA era. The analysis is primarily about generic market structure and does not focus on controlled-substance composition.

Controlled substances and opioid supply. Alpert, Powell, and Pacula (2018) show that the introduction of abuse-deterrent OxyContin reformulation in 2010 reduced prescription opioid misuse but was followed by substitution to heroin, illustrating the supply-side spillovers in controlled-substance markets. POWELL and PACULA (2021) extend this analysis to the fentanyl era. Neither paper studies the FDA approval side of the supply chain; this paper contributes by examining whether the upstream regulatory decision environment shaped the composition of what entered the market.

Gap. No existing study examines whether PDUFA or GDUFA changed the controlled-substance share of FDA approvals. The literature has studied review times, launch timing, post-market safety, and welfare under PDUFA, and market structure under GDUFA, but the composition question has not been addressed. This paper fills that gap.

3. Data

3.1. Sources

FDA Drugs@FDA. The primary data source is the Drugs@FDA flat-file database, downloaded from the FDA website in March 2026 (Research Fri, 03/13/2026 - 09:24). The database consists of twelve tab-delimited tables covering all FDA drug application and submission records from 1939 to the present. The core tables are `Applications` (one row per application, keyed by `ApplNo`), `Submissions` (one row per submission event, keyed by `ApplNo + SubmissionType + SubmissionNo`), and `Products` (product-level details linked to applications). Supporting tables cover submission classification codes, action types, document types, marketing status, and therapeutic equivalence codes.

I restrict attention to original approvals—applications with at least one submission classified as a new original application (`SubmissionClass` = ORIG or equivalent)—and collapse to one

row per application. This yields the drug-level panel of $N = 25{,}908$ observations used throughout the paper.

DEA controlled-substance reference. Controlled-substance status is determined by linking each FDA application to the DEA's official list of scheduled controlled substances (*Drug Scheduling* 2026). The linkage is ingredient-based: I match the active ingredient(s) associated with each FDA application to the DEA scheduling reference using a combination of exact-name matching, partial-string matching, and manual review. The match is conservative—only "confident scheduled" matches (ingredients with a clear one-to-one mapping to a DEA schedule) are classified as controlled substances in the primary analysis. Uncertain candidates and `List I` chemicals are excluded from the main CS indicator but appear in a broad-definition sensitivity check.

An important limitation of this approach is that the DEA linkage reflects *current* scheduling status, not historical status at the time of approval. Because scheduling can change over time, the current-status linkage may misclassify some drugs approved before they were scheduled and include others that were rescheduled after approval. The direction of the resulting bias is ambiguous but is likely small for the primary CS indicator, which focuses on commonly scheduled substances that have been on DEA schedules for decades.

3.2. Variable Construction

Outcome. The primary outcome is the annual share of original approvals that are controlled substances, constructed separately for the full sample, NDA-only, and ANDA-only subsamples. Let y_t denote the CS share in year t:

$$y_t = \frac{\text{CS approvals}_t}{\text{Total approvals}_t}$$

where "CS approvals" counts applications matched to a confident DEA schedule.

Application type. The variable `appltype_str` from the Drugs@FDA Applications table distinguishes NDA (new drug applications for brand-name and certain generic drugs), ANDA (abbreviated new drug applications for generic drugs), and BLA (biologics license applications). In the annual panel, 86.6% of CS approvals are ANDAs, reflecting the large volume of generic controlled-substance approvals—primarily opioid, benzodiazepine, and stimulant generics—processed through the ANDA pathway.

Policy timing. Three binary indicators capture the major policy transitions:

- `post_pdufa`: equals 1 for approval years 1993 and later.
- `post_gdufa`: equals 1 for approval years 2015 and later (the post-performance-goal era, excluding the 2013–2014 backlog-washout transition).
- `post_hatchwaxman`: equals 1 for approval years 1985 and later.

Event time. For event-study specifications, event time is measured relative to the policy year. For PDUFA, $\tau = t - 1992$, with the reference year set to $\tau = -1$ (1991). For GDUFA,

$\tau = t - 2012$, with reference year $\tau = -1$ (2011). End-caps are applied at $[-20, +30]$ for PDUFA and $[-10, +10]$ for GDUFA to limit sensitivity to sparse years far from the event.

3.3. Panel Structure

The drug-level panel covers 1939–2025. For the main event-study analysis, I aggregate to annual cells. The PDUFA specifications use the full 1939–2025 annual panel; the GDUFA specifications use the ANDA-only subsample covering 1984–2025. Stacked difference-in-differences specifications reshape the annual panel into a two-row-per-year format (CS group vs. non-CS group) as described in Section 4.

3.4. Summary Statistics

Table 1 reports the number of approvals and CS share by PDUFA reauthorization era. Several patterns stand out. First, the aggregate CS share is broadly stable in the 7–10% range across most eras, with a modest elevation in PDUFA II (2003–2007, 10.8%) and a sharp decline in PDUFA VII (2022–present, 4.5%). Second, ANDA volume grew dramatically after Hatch–Waxman (1984), accounting for the large increase in total approvals from the pre-Hatch–Waxman to post-Hatch–Waxman eras. Third, the NDA count has been relatively stable at 300–550 per five-year period across the PDUFA era.

Table 2 provides pre/post comparisons for PDUFA and GDUFA. The aggregate CS share fell from 9.6% pre-PDUFA to 7.3% post-PDUFA. Among NDAs, the CS share was nearly identical in the two periods (5.1% pre vs. 5.3% post). Among ANDAs, the CS share fell from 14.3% to 8.1%, driven largely by the post-Hatch-Waxman expansion in non-CS generic approvals. For GDUFA, the ANDA CS share fell from 10.8% (pre-GDUFA, $\leq$ 2012) to 7.1% (post-GDUFA performance-goal era, $\geq$ 2015).

Table 1: Original FDA Approvals by PDUFA Era

Era	Years	Total	NDA	ANDA	CS	CS Share
Pre-Hatch–Waxman	≤1984	4,270	1,488	2,005	392	9.2%
Post-HW, Pre-PDUFA	1985–1992	3,031	676	2,133	309	10.2%
PDUFA I	1993–1997	1,525	467	964	114	7.5%
PDUFA II	1998–2002	1,662	422	1,187	179	10.8%
PDUFA III	2003–2007	2,281	429	1,775	166	7.3%
PDUFA IV	2008–2012	2,761	394	2,300	210	7.6%
PDUFA V	2013–2017	3,464	551	2,830	268	7.7%
PDUFA VI	2018–2022	4,380	542	3,715	316	7.2%
PDUFA VII	2023+	2,534	308	2,113	113	4.5%
Total	1939–2025	25,908	4,277	19,022	2,067	8.0%

Source: Drugs@FDA (March 2026 extract) linked to DEA controlled-substance schedules.
CS = Controlled Substance (conservative: confident DEA scheduled matches only).
NDA counts exclude BLA (biologics) and unknown application types.

Table 2: Pre/Post Comparisons: PDUFA (1992) and GDUFA (2012)

	Total	NDA	ANDA	CS	CS/NDA	CS/ANDA
Panel A: PDUFA						
Pre-PDUFA (≤1992)	7,301	2,164	4,138	701	5.1%	14.3%
Post-PDUFA (≥1993)	18,607	3,113	14,884	1,366	5.3%	8.1%
Panel B: GDUFA (ANDA-only)						
Pre-GDUFA (≤2012)	8,359	—	8,359	858	—	10.8%
Post-GDUFA (≥2015)	7,830	—	7,830	556	—	7.1%

GDUFA transition years (2013–2014) are excluded from Panel B.
Pre-PDUFA period includes both pre-Hatch–Waxman and post-HW/pre-PDUFA years.

3.5. Limitations

Several data limitations bear on the interpretation of results. First, the DEA linkage reflects current scheduling status, not historical status at the time of approval. Second, the Drugs@FDA database is approval-centered: failed and withdrawn applications are not observed, which limits the ability to study the intensive margin of manufacturer drug development decisions. Third, the ingredient-level linkage may misclassify combination products with both controlled and non-controlled ingredients; the variable `is_multi_ingredient` flags these cases (8% of CS approvals). Fourth, review-priority coding (`review_priority`) is available but unstable over time and is not used as a main outcome. Fifth, the annual series is a single national time series without cross-sectional variation in the main specifications, limiting the credibility of causal inference.

4. Empirical Strategy

4.1. Research Design

The ideal experiment would randomly assign PDUFA or GDUFA implementation across otherwise identical regulatory environments and observe the resulting change in the controlled-substance share of approvals. In practice, both policies were enacted at single national dates and applied to all drug approvals simultaneously, so there is no cross-sectional control group in the standard sense. The analysis therefore relies on time-series comparisons—interrupted time series (ITS) and difference-in-differences (DD) designs—with pre-policy trends serving as the counterfactual for the post-policy period. I treat the estimates as policy-diagnostic first-pass evidence rather than as point-identified causal effects.

4.2. Interrupted Time Series (ITS)

The primary specification is a linear ITS estimated on the annual CS-share series:

$$y_t = \alpha + \beta_1 \cdot \text{post}_t + \beta_2 \cdot t + \beta_3 \cdot (\text{post}_t \times t) + \varepsilon_t \tag{1}$$

where y_t is the annual CS share in year t, post_t is the post-policy indicator, and the interaction $\text{post}_t \times t$ allows the post-policy trend to differ from the pre-policy trend. The coefficient β_1

estimates the level shift at the policy date; β_3 estimates the change in slope. The ITS is estimated separately for PDUFA (1993) and GDUFA (2015). For PDUFA, the sample is restricted to 1970–2025 to focus on the relevant policy window; for GDUFA, the ANDA-only series is used and 2013–2014 are excluded to isolate the backlog-washout transition.

4.3. Poisson Rate Model

Because the annual CS share is a proportion bounded between 0 and 1 and the numerator (count of CS approvals) is non-negative, the linear probability model in equation (1) may fit poorly in eras with very low or very high shares. I therefore also estimate a Poisson regression treating the annual CS count as the dependent variable and the total annual approval count as an exposure offset:

$$\log E[\mathrm{CS}_t] = \alpha + \gamma \cdot \mathrm{post}_t + \log(\mathrm{Total}_t) \tag{2}$$

where $\exp(\hat{\gamma})$ is the incidence-rate ratio (IRR) for the post-policy period relative to pre-policy, holding the total approval volume constant. Equation (2) and its variants add year fixed effects or linear time trends to absorb secular changes in the composition of approvals. All Poisson models use heteroskedasticity-robust standard errors.

4.4. Stacked Difference-in-Differences

To increase transparency about the shape of the post-policy change, I also estimate a stacked difference-in-differences specification. For each policy, I reshape the annual panel into a two-row-per-year dataset in which one observation represents CS approvals and the other represents non-CS approvals in each year:

$$\log E[n_{gt}] = \alpha_g + \lambda_t + \delta \cdot (\mathrm{CS}_g \times \mathrm{post}_t) + \log(\mathrm{Total}_t) \tag{3}$$

where $g \in \{\mathrm{CS}, \text{non-CS}\}$ indexes the group, α_g is a group fixed effect, λ_t is a year fixed effect, and δ is the DD coefficient of interest. The post indicator post_t drops out of the year-FE specification by collinearity. The Poisson DD is estimated with exposure Total_t; $\exp(\hat{\delta})$ is the IRR comparing the relative change in CS approvals to non-CS approvals after the policy. Linear (levels) and inverse hyperbolic sine (IHS) specifications are also estimated for robustness.

4.5. Event-Study DD

To examine the dynamics of any post-policy change and test for pre-policy parallel trends, I estimate an event-study version of the DD:

$$\log E[n_{g\tau}] = \alpha_g + \sum_{\tau \neq -1} \theta_\tau \cdot (\mathrm{CS}_g \times \mathbf{1}[\tau]) + \log(\mathrm{Total}_{g\tau}) \tag{4}$$

where τ is event time (calendar year relative to the policy date), and θ_τ is the group-specific coefficient at each event-time bin. End-caps are applied at $\tau \in [-20, +30]$ for PDUFA and $\tau \in [-10, +10]$ for GDUFA. The reference period is $\tau = -1$ (the year before each policy). Under

the parallel-trends assumption, $\theta_\tau = 0$ for all $\tau < 0$; a pattern of pre-policy zeros followed by post-policy deviations from zero would constitute evidence of a treatment effect (Miller 2023).

4.6. Threats to Identification

Three threats to identification are particularly relevant in this context.

Confounders contemporaneous with policy. Both PDUFA (1992) and GDUFA (2012) were enacted at times when other forces were simultaneously reshaping the pharmaceutical market. PDUFA coincided with a period of rapid expansion in ANDA volume following Hatch–Waxman, which could confound any composition effect by diluting non-CS approvals. GDUFA coincided with a surge in generic applications filed to capture expiring patents from the 2005–2012 "patent cliff." I address this by using application-type-specific subsamples (NDA-only for PDUFA, ANDA-only for GDUFA) and by including year fixed effects where applicable.

Opioid product cycle (2009–2013). A spike in Schedule II NDA approvals around 2009–2013 is visible in the data and drives the apparent post-PDUFA effect in some specifications. Section 5.3 and Section 6 document that this spike reflects a specific industry product cycle—the wave of extended-release and abuse-deterrent branded opioid formulations (OxyContin reformulation, Exalgo, Embeda, Suboxone)—rather than a sustained PDUFA-induced composition shift. The event-study design is helpful for diagnosing this: a genuine PDUFA effect should appear near $\tau = 0$ (1992–1993), not 17–19 years later.

Single time series. The national-level panel has only one observation per year, so the effective sample size for time-series inference is small (roughly 50–85 years depending on the specification). Standard errors based on heteroskedasticity alone may understate uncertainty; inference should be treated with appropriate caution.

4.7. What the Design Can and Cannot Show

This design can characterize the *reduced-form association* between PDUFA/GDUFA and the controlled-substance share of approvals. A finding of no effect would be informative: it would indicate that user-fee-funded regulatory acceleration did not systematically tilt the composition of approvals toward controlled substances. A finding of a positive effect would be consistent with the regulatory-incentive hypothesis but would not rule out confounding by contemporaneous trends.

What the design *cannot* show is whether any composition change affected downstream misuse, diversion, or overdose mortality. That would require linking the approval-side estimates to prescribing data and ultimately to health outcomes—a step outside the scope of this paper. The contribution is upstream: characterizing the approval-side composition effect (or lack thereof) is a necessary first step before studying downstream consequences.

5. Results

5.1. Descriptive Trends

Figure 1 shows the annual share of original FDA approvals classified as controlled substances from 1939 to 2025. Three features stand out. First, the share is broadly stable in the 5–15% range over the full sample, without a clear structural break at PDUFA (1992). Second, the share exhibits substantial year-to-year noise in the early part of the sample when annual approval counts were small. Third, a moderate spike is visible in the 2009–2012 period, which is investigated in Section 5.5.

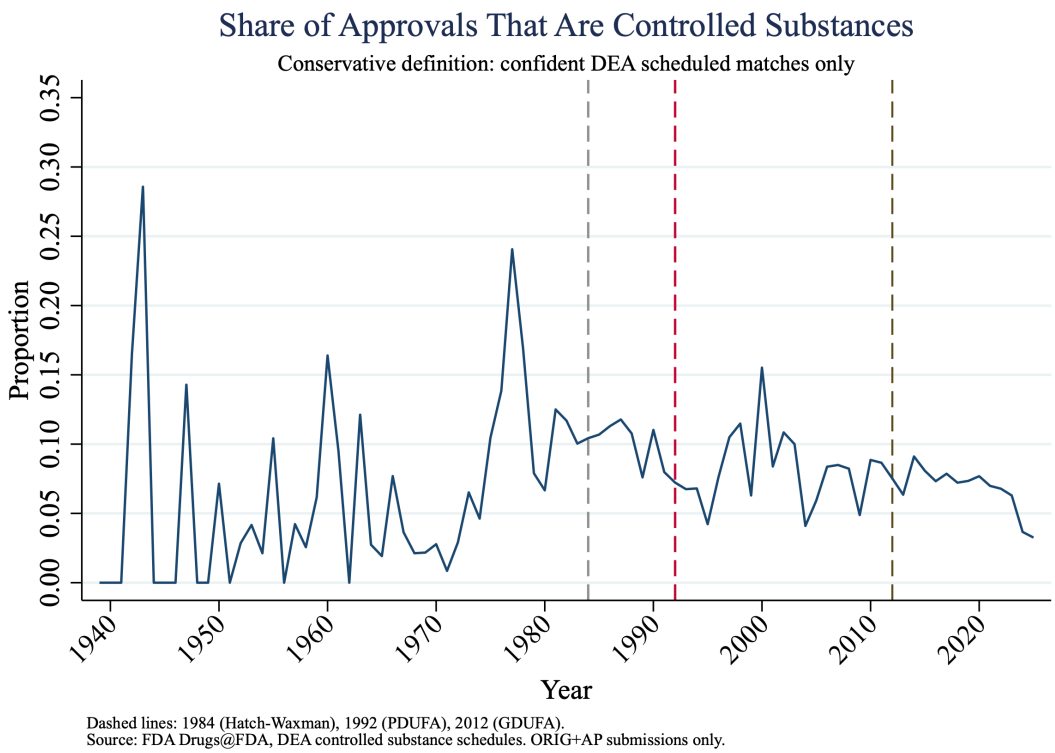

Figure 1: Annual controlled-substance share of FDA original approvals, 1939–2025. Conservative definition: confident DEA scheduled matches only. Dashed vertical lines mark 1984 (Hatch–Waxman), 1992 (PDUFA), and 2012 (GDUFA).

Figure 2 disaggregates the CS share by application type (NDA vs. ANDA). The NDA CS share fluctuates narrowly around 5–8% throughout the sample with no visible trend break at PDUFA. The ANDA CS share is higher—peaking near 20% in the 1980s—and declines after Hatch–Waxman as the expansion in non-CS generic approvals dilutes the CS share. Years with fewer than five NDA or ANDA approvals are excluded to suppress noise.

5.2. PDUFA Event Study: Full Sample

Figure 3 presents the event-study point estimates and 95% confidence intervals from the ITS specification (equation 1), with event time measured in years relative to PDUFA (1992). The pre-policy estimates ($\tau < 0$) are clustered near zero, consistent with the parallel-trends assumption.

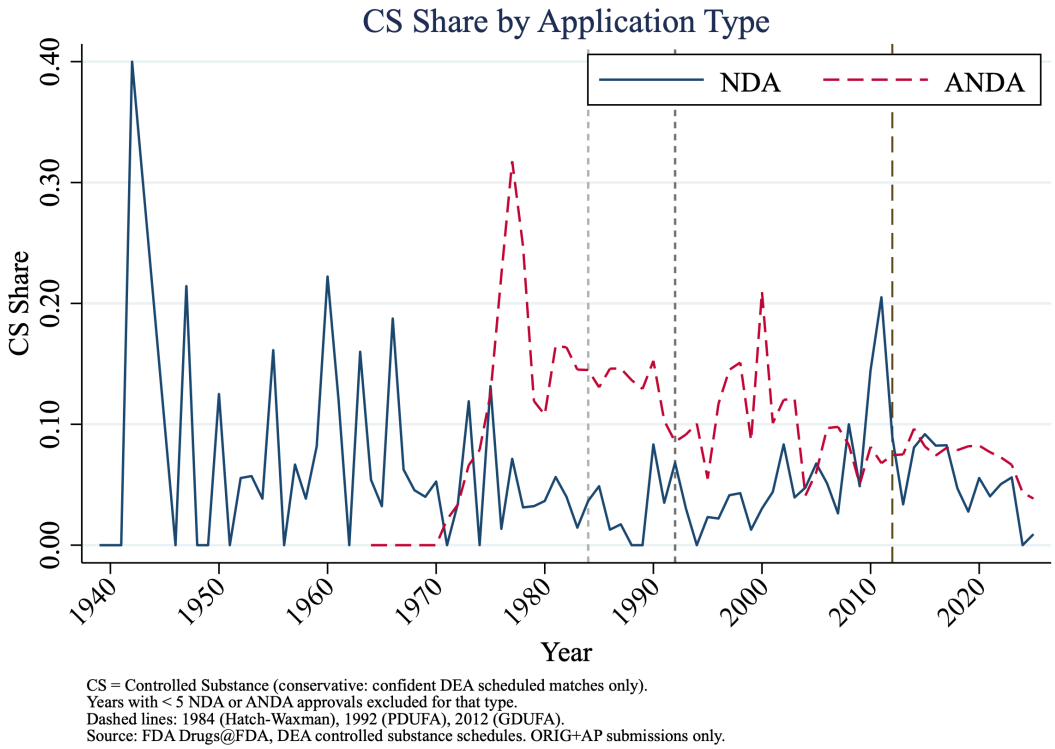

Figure 2: Annual CS share by application type (NDA vs. ANDA), 1939–2025. Years with fewer than 5 NDA or ANDA approvals are excluded. Dashed lines as in Figure 1.

Post-policy estimates ($\tau > 0$) show modest, statistically insignificant deviation from the pre-trend. There is no visible level shift or trend break at $\tau = 0$. The wide confidence intervals in both the pre- and post-periods reflect the limited power of a single national time series.

5.3. PDUFA Event Study: NDA-Only Sample

Figure 4 restricts the event-study to new drug applications (NDAs). This is the theoretically relevant subsample: PDUFA directly governed NDA review times and does not apply to ANDA submissions. The NDA CS share series is noisier because annual NDA counts are smaller, but the event-study pattern shows an important feature: the largest positive deviations from the pre-trend occur around event times 17–19 (calendar years 2009–2011), not at or near $\tau = 0$. A genuine PDUFA effect on NDA composition should be visible near the policy date.

Figure 5 shows the NDA CS share level series with pre-PDUFA and post-PDUFA mean reference lines. The post-PDUFA mean is marginally higher than the pre-PDUFA mean, but the time-series visual makes clear that the elevation is driven by the 2009–2013 period.

Table 3 reports Poisson rate models for the NDA CS count. Model (1) uses year fixed effects; Model (2) replaces year FEs with a linear time trend; Model (3) adds group-specific trends. The IRR in Model (1) is 1.46 ($p = 0.060$), suggesting a 46% higher CS approval rate post-PDUFA relative to pre-PDUFA after absorbing year-level variation. However, Model (3)—which allows pre- and post-PDUFA trends to differ by group—yields an IRR of 1.19 ($p = 0.62$), indicating that the apparent effect in Model (1) is substantially explained by differential pre- versus post-policy

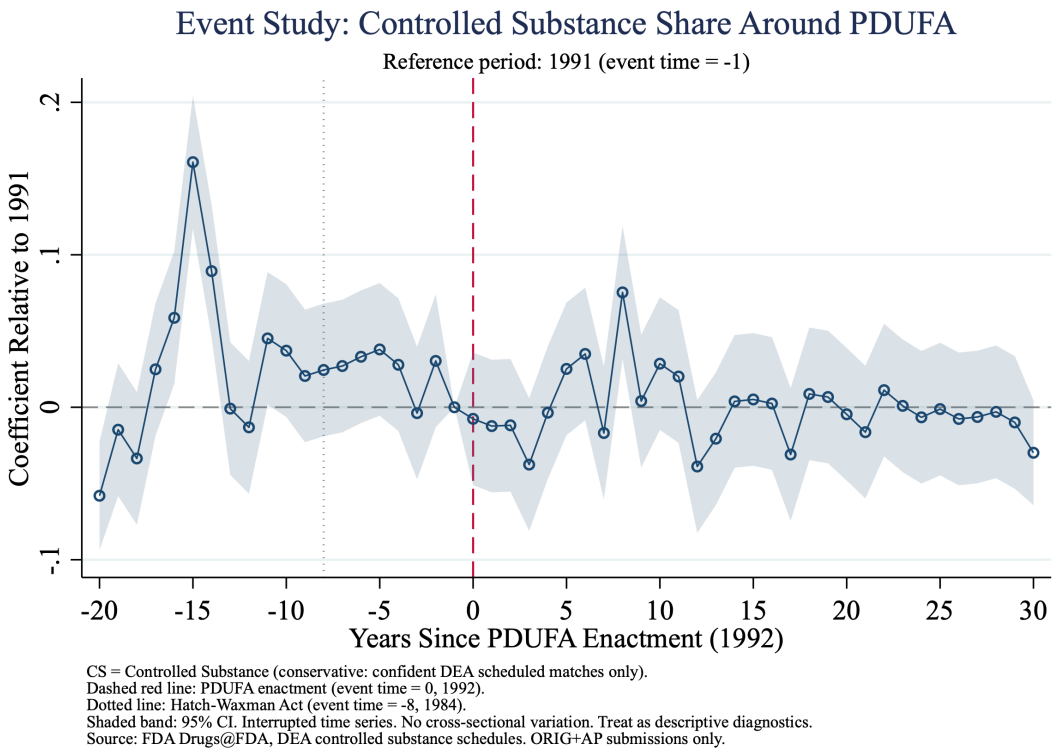

Figure 3: PDUFA event study: ITS coefficients on CS share. Event time 0 = 1992 (PDUFA enactment). Reference period: $\tau = -1$ (1991). End-caps at $[-20, +30]$. 95% confidence intervals.

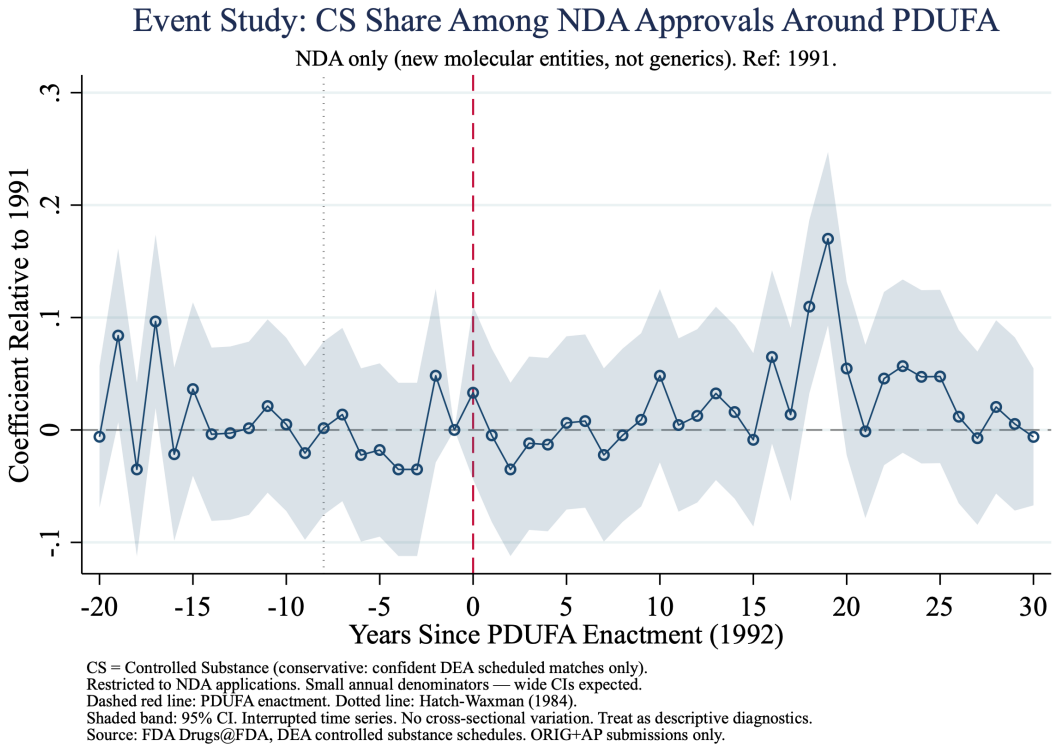

Figure 4: PDUFA event study: NDA-only sample. CS share among new drug applications. Event time 0 = 1992. Reference: $\tau = -1$.

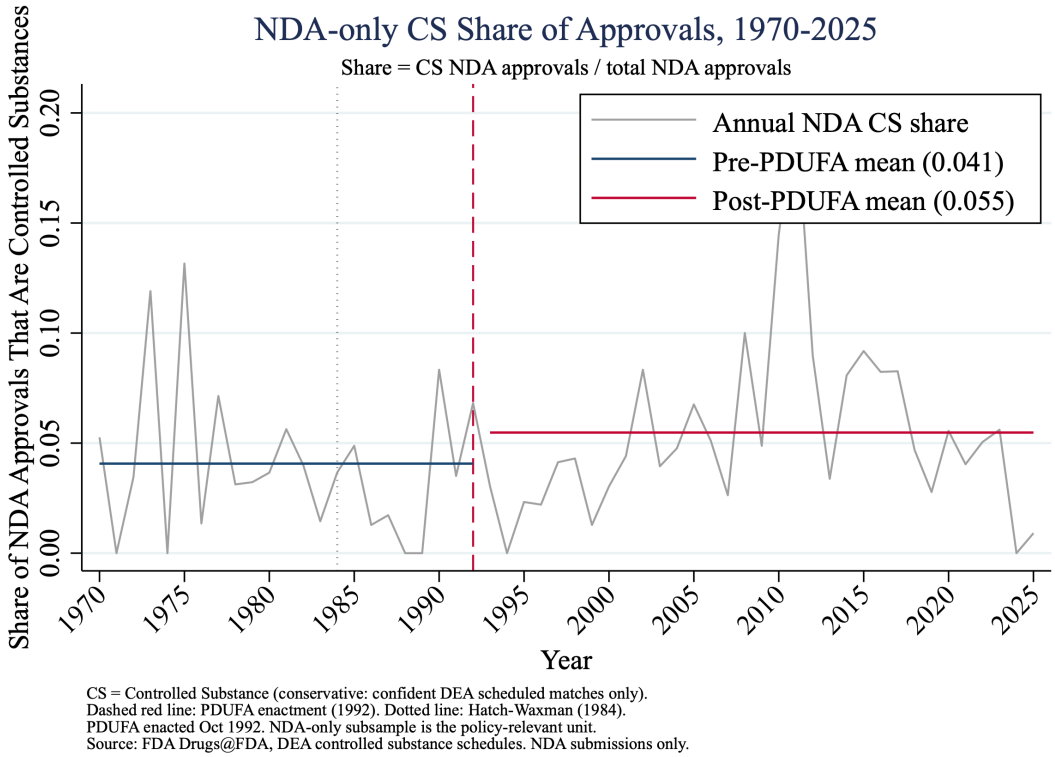

Figure 5: NDA controlled-substance share, 1970–2025, with pre/post-PDUFA mean reference lines.

trends rather than a level shift at PDUFA.

Table 3: PDUFA NDA-Only Poisson Rate Models: Sensitivity to Trend Controls

	Post-PDUFA IRR (CS vs. non-CS)		
	(1) Year FE	(2) Linear trend	(3) Group trends
IRR	1.456	1.456	1.190
z-statistic	1.879	1.832	0.490
p-value	0.060	0.067	0.624
Observations	Annual, 1970–2025, NDA only		
Exposure	Total NDA approvals		

IRR = incidence-rate ratio for post_pdufa × is_cs interaction. Robust standard errors.
Model (3) adds separate pre-policy and post-policy linear time trends by group.

Figure 6 shows the event-study DD coefficients for the NDA-only stacked Poisson specification. The IRR profile is flat and near zero for all pre-policy event times, consistent with parallel pre-trends. Post-policy IRRs drift upward gradually, reaching their largest values around event times 17–19 (2009–2011), before subsiding. This pattern—a late-rising post-policy departure that does not appear near the policy date—is inconsistent with a PDUFA effect and is instead consistent with a delayed industry product cycle, as investigated in Section 5.5.

5.4. Rate Versus Count Decomposition

Figure 7 presents a two-panel decomposition of the NDA CS result. Panel A shows the CS share of NDA approvals (the rate); Panel B shows the absolute counts of CS NDA approvals and non-CS NDA approvals over the same period. The panels illustrate that the NDA CS rate rises modestly after 1992 even as the absolute volume of non-CS NDA approvals grows faster in levels. Both patterns are simultaneously consistent: the denominator (non-CS NDA volume) growing faster than the numerator (CS NDA volume) in absolute terms does not preclude the CS rate from rising if CS NDAs grow faster in proportional terms relative to the pre-period baseline.

5.5. The 2009–2013 CS NDA Spike

The event-study and time-series evidence point to a spike in Schedule II NDA approvals concentrated in 2009–2013. Table 4 lists the CS NDA approvals from 2009 through 2013, identifying the brand name, active ingredient, sponsor, and DEA schedule for each drug. Four features stand out. First, 43 CS NDAs were approved in this five-year window, compared with an average of roughly 6–8 per year in the preceding decade. Second, 2011 alone accounts for 16 CS NDA approvals—by far the largest single-year count in the sample. Third, 25 of the 43 approvals (58%) are Schedule II, the most restrictive category for drugs with accepted medical use. Fourth, the drug list maps closely to the well-documented wave of branded extended-release and abuse-deterrent opioid formulations that characterized the late-2000s pharmaceutical market: OxyContin reformulation (Purdue Pharma, 2010), Exalgo hydromorphone (Specgx, 2010),

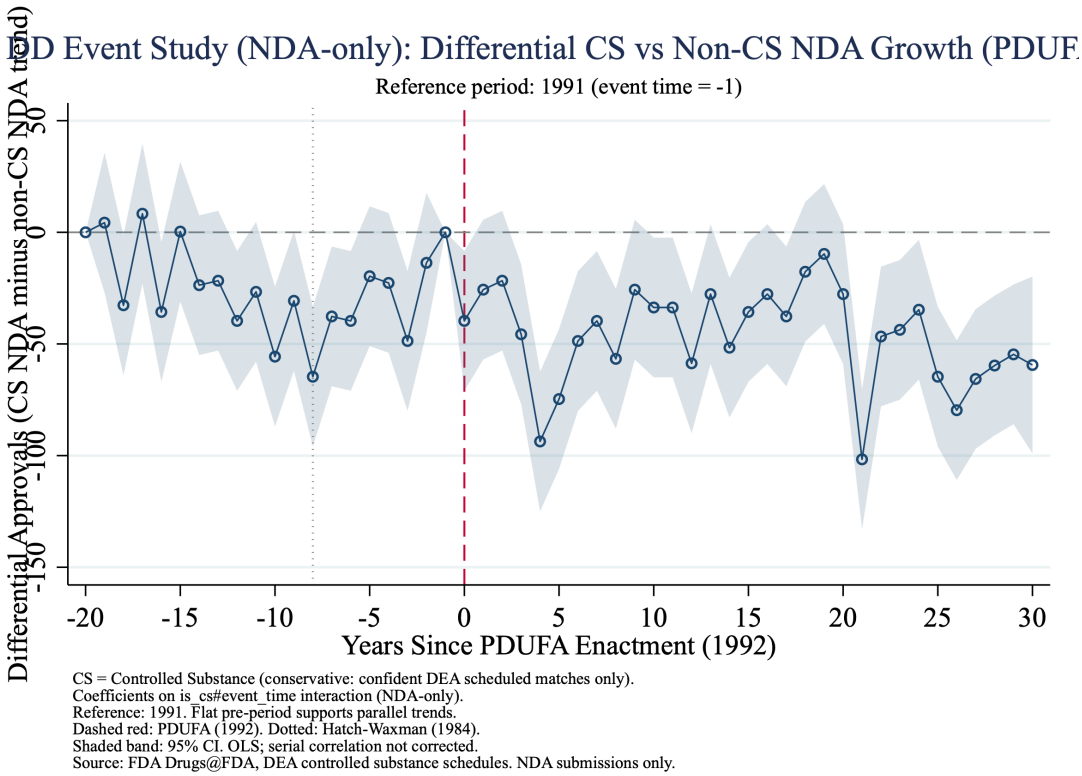

Figure 6: PDUFA NDA-only stacked Poisson DD event study. IRRs for CS $\times$ $\mathbf{1}[\tau]$ interactions. Reference: $\tau = -1$ (1991). End-caps at $[-20, +30]$. 95% CI.

Embeda morphine-naltrexone combination (Alpharma, 2009), Suboxone buprenorphine-naloxone (Indivior, 2010), and related products.

5.6. GDUFA Event Study: ANDA-Only

Figure 8 presents the GDUFA event-study DD for the ANDA-only sample. In contrast to the PDUFA result, the GDUFA event study shows a pronounced and statistically significant decline in the CS share of ANDA approvals after the post-performance-goal era begins (event times 3–13, corresponding to calendar years 2015–2025). Pre-policy event-time coefficients are clustered near zero, supporting the parallel-trends assumption over the 2002–2012 pre-GDUFA window. The stacked Poisson DD specification (Table 5) yields an IRR of 0.668 ($p < 0.001$) under year fixed effects, indicating that the CS share of ANDA approvals fell by approximately 33% relative to the non-CS share after GDUFA. This result is also robust to a linear time-trend control (IRR $= 0.668$, $p = 0.008$), though the group-specific trend specification yields an insignificant IRR of 1.38 ($p = 0.18$).

Figure 9 decomposes the GDUFA ANDA result into rate (Panel A) and count (Panel B). Panel A shows the ANDA CS share declining from a pre-GDUFA mean of 10.8% to a post-GDUFA mean of approximately 7.1%. Panel B shows that both the CS ANDA count and non-CS ANDA count declined in absolute terms after GDUFA, with the CS count falling proportionally more steeply.

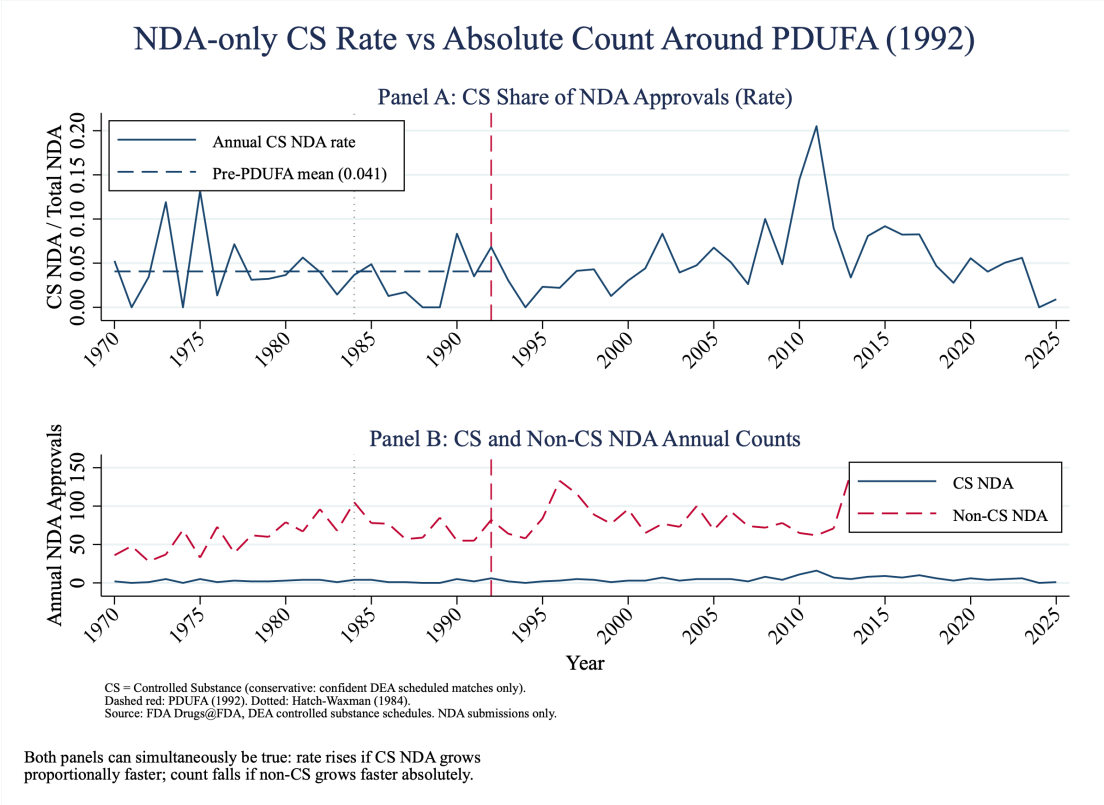

Figure 7: NDA CS approvals: rate (Panel A) and absolute count (Panel B). Panel A shows CS share among NDA approvals with pre-PDUFA mean reference line. Panel B shows annual CS NDA count vs. non-CS NDA count.

Table 4: CS NDA Approvals, 2009–2013 (Selected)

Year	Brand Name	Active Ingredient	Sponsor	Sched.
2009	ONSOLIS	Fentanyl citrate	ADALVO	II
2009	EMBEDA	Morphine sulfate; naltrexone	ALPHARMA	II
2009	CODEINE SULFATE	Codeine sulfate	HIKMA	II
2009	EDLUAR	Zolpidem tartrate	VIATRIS	IV
2010	EXALGO	Hydromorphone HCl	SPECGX LLC	II
2010	OXYCONTIN	Oxycodone HCl	PURDUE PHARMA	II
2010	SUBOXONE	Buprenorphine; naloxone	INDIVIOR	III
2010	BUTRANS	Buprenorphine	PURDUE PHARMA	III
2010	AXIRON	Testosterone	ELI LILLY	III
2010	LYRICA	Pregabalin	UPJOHN	V
2011	ANDROGEL	Testosterone	BESINS HLTHCARE	III
2011	INTERMEZZO	Zolpidem tartrate	PURDUE PHARMA	IV
2012	ZOHYDRO ER	Hydrocodone bitartrate	ZOGENIX	II
Full 2009–2013 window: 43 CS NDAs; 25 Schedule II (58%); 2011 peak: 16 approvals.				

Selected rows for illustration; full list in `output/tables/stata/cs_nda_2009_2013_drugs.csv`.
OxyContin 2010 entry is a reformulated abuse-deterrent NDA (NDA 022272), not the original approval.

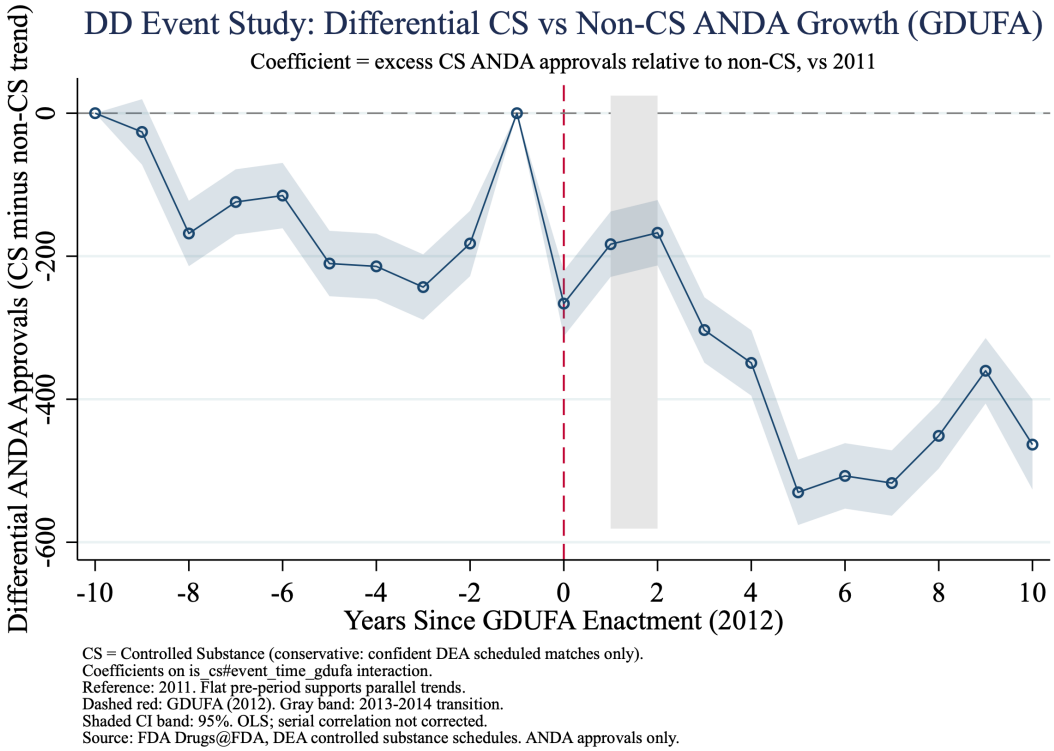

Figure 8: GDUFA ANDA-only stacked Poisson DD event study. IRRs for CS $\times$ $\mathbf{1}[\tau]$ interactions. Reference: $\tau = -1$ (2011). End-caps at $[-10, +10]$. Transition years 2013–2014 excluded. 95% CI.

Table 5: GDUFA and PDUFA Stacked Poisson DD: Main Results

	Coef.	SE	IRR	p
Panel A: GDUFA, ANDA-only (Year FE)				
post_gdufa $\times$ is_cs	-0.403	0.095	0.668	<0.001
Panel B: PDUFA, NDA-only (Year FE)				
post_pdufa $\times$ is_cs	0.376	0.200	1.456	0.060

Stacked Poisson DD with year FE and log-total-approval exposure. Robust SE.
GDUFA excludes 2013–2014 transition. PDUFA uses NDA-only sample.

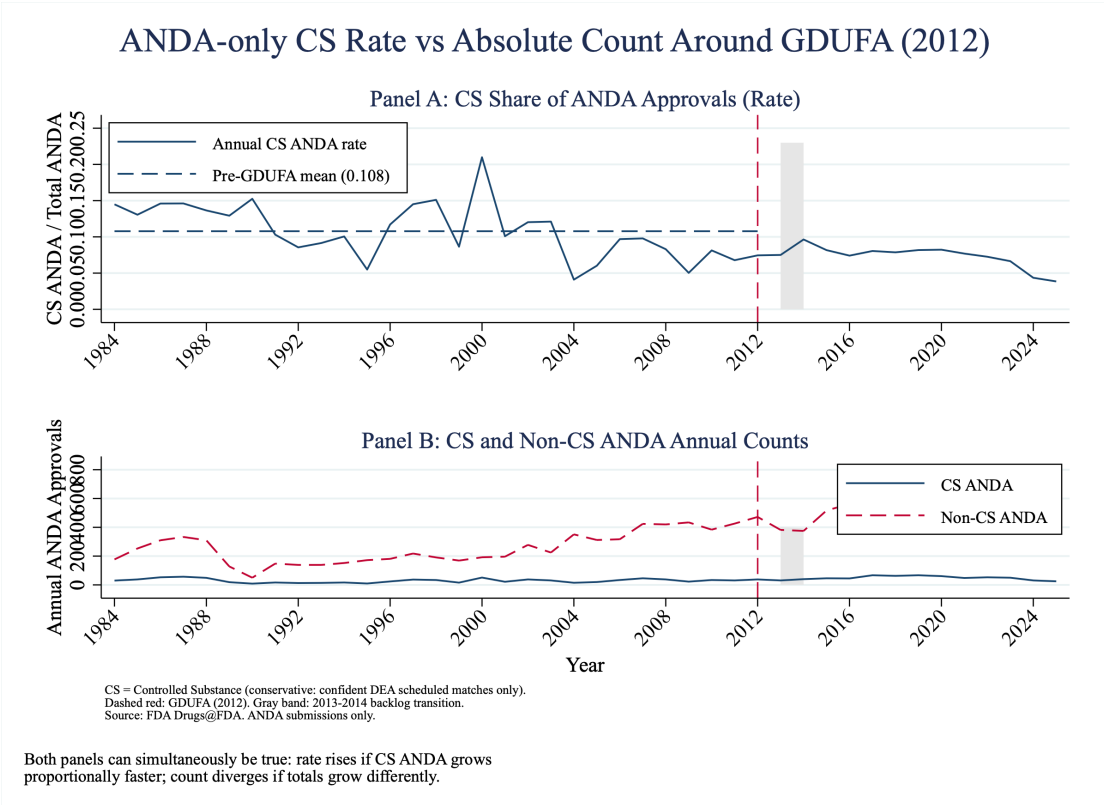

Figure 9: GDUFA ANDA CS approvals: rate (Panel A) and absolute count (Panel B). Shaded region: 2013–2014 transition period.

5.7. Sponsor Concentration

Figure 10 and Figure 11 show the Herfindahl–Hirschman Index (HHI) of sponsor concentration among CS NDA approvals (Figure 10) and all ANDA approvals (Figure 11), respectively. The NDA CS HHI is notably higher and more variable than the ANDA HHI, reflecting the smaller number of annual CS NDA approvals and the dominance of a small number of branded opioid sponsors in certain years. The GDUFA era is associated with a visible decline in ANDA CS HHI, consistent with both the entry of additional generic sponsors and the decline in CS ANDA volume.

Table 6 reports the top 10 sponsors of CS NDA approvals over the full 1970–2025 period. The market is fragmented: the leading sponsor (HIKMA and UCB INC, tied at 9 CS NDAs each) accounts for only 4.1% of the 222 CS NDAs in the 1970–2025 window. Purdue Pharma LP ranks fourth with 7 approvals (3.2%), reflecting its historical dominance in branded opioid formulations but also the relatively small absolute scale of any single sponsor's CS NDA portfolio.

6. Discussion

6.1. Interpreting the Null for PDUFA

The aggregate evidence does not support the hypothesis that PDUFA caused a sustained increase in the controlled-substance share of FDA drug approvals. The raw CS share fell after 1992. The NDA-only Poisson rate model yields a marginally significant IRR of 1.46 under year fixed

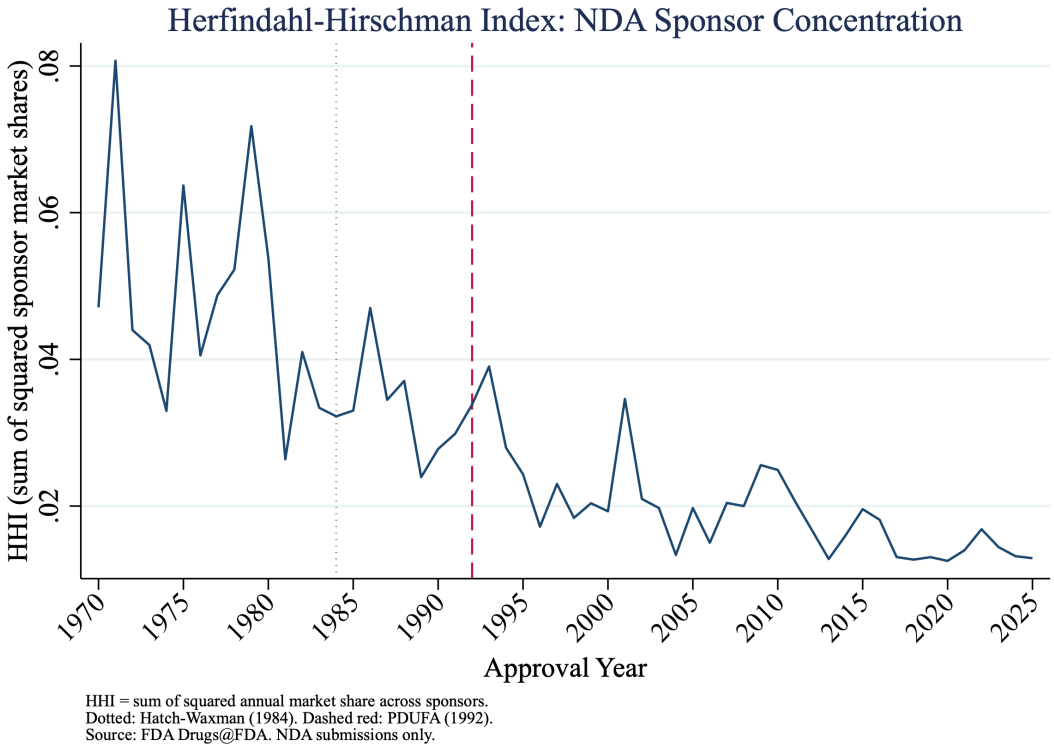

Figure 10: Sponsor HHI for CS NDA approvals, 1970–2025. Higher HHI = greater sponsor concentration. PDUFA reference lines marked.

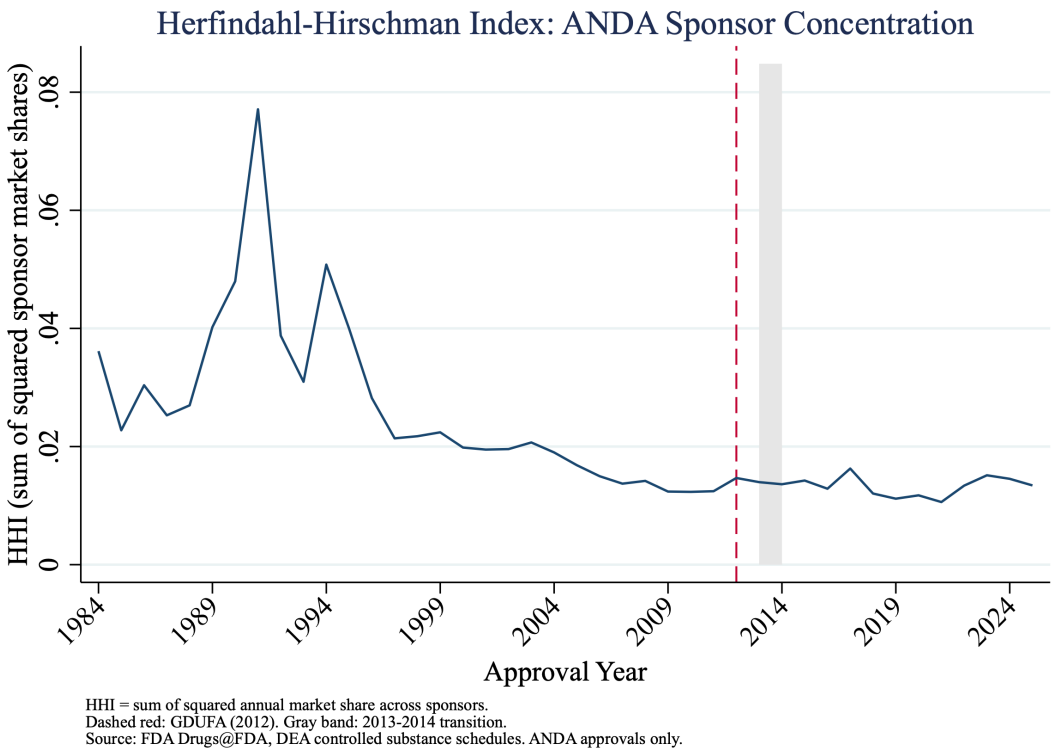

Figure 11: Sponsor HHI for ANDA approvals (all), 1984–2025. GDUFA reference line marked.

Table 6: Top 10 Sponsors of CS NDA Approvals, 1970–2025

Rank	Sponsor	CS NDAs	Share (%)
1	HIKMA	9	4.05
1	UCB INC	9	4.05
3	SPECGX LLC	7	3.15
3	PURDUE PHARMA LP	7	3.15
5	INDIVIOR	5	2.25
5	SANOFI AVENTIS US	5	2.25
5	UPJOHN	5	2.25
8	ABBOTT	4	1.80
8	COLLEGIUM PHARM INC	4	1.80
8	JANSSEN PHARMS	4	1.80
8	TAKEDA PHARMS USA	4	1.80

Universe: CS NDAs with confident DEA schedule match, 1970–2025, sponsor name non-missing.
Total CS NDAs in window: 222. Source: `cs_nda_top_sponsors_1970_2025.csv`.

effects, but this estimate is not robust: it collapses to 1.19 ($p = 0.62$) once group-specific pre- and post-policy trends are allowed. The event-study profile shows no shift at $\tau = 0$; the largest post-policy deviation occurs at $\tau \approx 17$–19 (2009–2011), far from the policy date.

These patterns are consistent with at least two non-PDUFA explanations. First, the post-Hatch-Waxman expansion in generic (ANDA) approvals after 1985 generated a large and growing pool of non-CS approvals that mechanically diluted the CS share of total approvals even if the absolute number of CS approvals continued to grow. The NDA-only restriction corrects for this composition effect at the application-type level, but even within NDAs, secular trends in the relative attractiveness of different therapeutic areas may confound inference. Second, the rise and fall of opioid pharmaceutical innovation over the 2000s and 2010s generated a specific industry product cycle that is visible in the data but is not plausibly attributable to PDUFA incentives operating 17 years later.

6.2. The 2009–2013 Opioid Product Cycle as a Separable Phenomenon

The drug-level evidence in Table 4 makes a strong case that the 2009–2013 CS NDA spike reflects the branded opioid extended-release and abuse-deterrent wave, not a PDUFA composition effect. The key features are: (1) the timing—18–21 years after PDUFA enactment—is inconsistent with an immediate regulatory incentive effect; (2) the drugs themselves are recognizable products from the documented opioid product innovation cycle, including reformulated OxyContin, Exalgo, Embeda, Suboxone, and multiple testosterone formulations; and (3) the spike subsides after 2013, suggesting a product-cycle dynamic rather than a permanent shift in approval composition.

The appropriate interpretation of this spike, for this paper's purposes, is that it is a confound: it inflates post-PDUFA IRRs in specifications without group-specific trends because the pre-PDUFA NDA CS rate was low and the spike creates a large mean difference in the post period. The fact that the group-specific-trend specification eliminates significance confirms that the spike is driving the result.

This finding is consistent with the broader opioid supply literature. Alpert, Powell, and Pacula

(2018) document that the branded opioid wave was followed by a substitution response in illicit markets; the approval-side documentation here provides complementary upstream evidence. The implication for policy is that the regulatory approval channel and the industry innovation channel are distinct: FDA approval decisions may have facilitated the opioid wave by allowing these products to come to market, but the primary driver of the wave appears to be firm-level pharmaceutical investment decisions, not the PDUFA review-time incentive structure.

6.3. Interpreting the GDUFA Decline

The GDUFA result—a 33% decline in the ANDA CS share post-2015—is the paper's most precisely estimated finding. The direction is opposite to the regulatory-acceleration hypothesis: if faster review disproportionately benefits CS generics, we would expect the CS ANDA share to rise after GDUFA. The fact that it falls is consistent with several alternative explanations.

First, the pre-GDUFA ANDA backlog may have disproportionately included CS generics filed during the 1990s–2000s opioid expansion. Clearing the backlog during 2013–2014 would have resulted in elevated CS ANDA approvals in the transition period and a subsequent reversion as the backlog drained. The three-period design (pre, transition, post-goals) partially accounts for this, but the transition-year exclusion may not fully isolate the pre-existing stock effect.

Second, the composition of ANDA applications submitted after GDUFA may have shifted toward non-CS generics, reflecting the 2012 patent cliff for non-opioid branded drugs and the subsequent filing pipeline. If non-CS generic applications surged after 2012 and began receiving approvals after the performance-goal era began in 2015, the CS share would mechanically fall even without any change in CS-specific review time or approval rates.

Third, the prescription opioid market contracted significantly after 2012 as prescribing guidelines tightened, state prescription drug monitoring programs expanded, and supply-side interventions restricted access. Reduced demand for opioid generics may have discouraged new ANDA filings for CS products relative to non-CS products.

The group-specific-trend specification yields an insignificant IRR of 1.38 for GDUFA, suggesting the same sensitivity to trend assumptions as the PDUFA result. This cautions against strong causal claims in either direction.

6.4. Limitations and Future Work

Several limitations constrain the conclusions. The most significant is the DEA linkage: because current schedules are used rather than historical schedules, any drugs that were scheduled or rescheduled after their FDA approval date are misclassified. The direction of this error for the main results is unclear. A second limitation is the single national time series, which provides limited statistical power and cannot separate PDUFA effects from contemporaneous secular trends in pharmaceutical innovation. A third limitation is the absence of a downstream outcome: this paper characterizes the approval-side composition of CS drugs but does not measure downstream prescribing volume, misuse, or diversion.

Directions for future work include: (1) constructing a historical DEA scheduling database to determine each drug's schedule as of its approval date; (2) linking approval-side CS counts to prescribing data from IQVIA or ARCOS to estimate the downstream supply implications of any

composition change; (3) extending the GDUFA analysis to GDUFA II and III as those program years accumulate; and (4) exploiting therapeutic-class or firm-level variation to identify the mechanism—review-time incentives versus aggregate staffing capacity—driving any composition effect.

7. Conclusion

This paper asks whether PDUFA (1992) and GDUFA (2012)—the two major user-fee-funded regulatory acceleration reforms at the FDA—changed the controlled-substance share of original drug approvals. Using the complete Drugs@FDA database (1939–2025, $N = 25{,}908$) linked to DEA scheduling records, I estimate interrupted time series and stacked difference-in-differences specifications for both the NDA pathway (PDUFA) and the ANDA pathway (GDUFA).

The central finding is an empirical null for PDUFA and a direction-reversing result for GDUFA. The aggregate CS share of FDA approvals fell after 1992. The NDA-only Poisson rate model yields an IRR of 1.46 that is marginally significant under year fixed effects but not robust to group-specific trend controls. A drug-level investigation confirms that the apparent post-PDUFA elevation in CS NDA approvals is driven by a specific 2009–2013 industry product cycle—extended-release and abuse-deterrent branded opioid formulations—rather than by a sustained PDUFA-induced composition shift. For GDUFA, the ANDA CS share fell significantly in the post-performance-goal era (IRR $= 0.67$, $p < 0.001$), inconsistent with the hypothesis that faster generic review disproportionately benefited CS products.

The findings suggest that the standard economic argument for why user-fee-funded regulatory acceleration should raise the CS share of approvals—that CS drugs have higher expected margins and therefore more to gain from faster review—does not find support in the approval data. The controlled-substance composition of FDA approvals appears to be driven by industry product cycles and application-mix trends rather than by review-time incentives at the regulatory margin. This does not imply that FDA regulatory policy is irrelevant to controlled-substance supply: approval decisions are a necessary condition for market entry, and any drug on the list in Table 4 required FDA clearance before it could be prescribed. But the compositional variation is better explained by demand for specific formulations in the pharmaceutical product cycle than by PDUFA incentives.

For researchers and policymakers, the implication is that the causal pathway from regulatory acceleration to controlled-substance supply likely runs primarily through the innovation investment channel—firms deciding whether to develop CS products—rather than through the review-time channel. Understanding that investment decision, and how factors like patent term, market exclusivity, and prescribing-market demand conditions shape it, may be more productive for policy design than focusing on FDA review-time performance goals. Future work linking the upstream approval-side evidence here to downstream prescribing volumes and ultimately to health outcomes would provide a more complete picture of how pharmaceutical regulation shapes controlled-substance supply across the full market chain.

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A. Appendix

Open TODOs

No open TODOs.